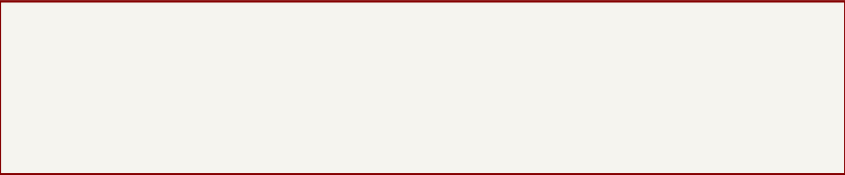


BOOT CONSOLE

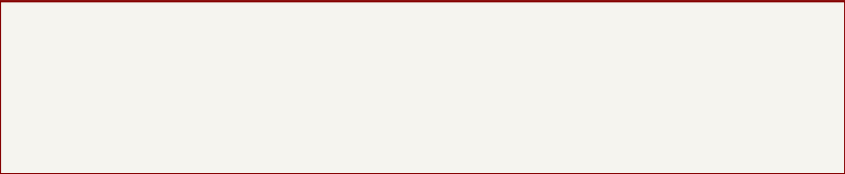
F1C200s / 64 MiB SiP DDR1 / microSD Linux / USB-C / UART0

01 / Power and supervised startup



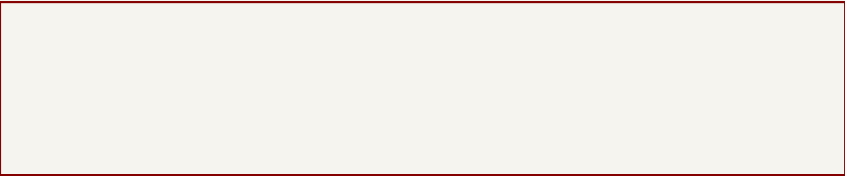
File: power.kicad_sch

02 / F1C200s, clock and local bypass



File: processor.kicad_sch

03 / USB-C, microSD and UART console



File: interfaces.kicad_sch

Mounting holes / M2.5

H1
M2.5 / 2.7mm



H2
M2.5 / 2.7mm



H3
M2.5 / 2.7mm



H4
M2.5 / 2.7mm



USB-C

5 V power + FEL recovery.
USB serial requires a Linux gadget configuration.

UART0

3.3 V logic / 115200 8N1
J3: GND, board TX, board RX

MECHANICAL

44 x 44 mm, R3 corners
Four 2.7 mm M2.5 clearance holes
Black mask / white silk / ENIG
0603 passives; larger bulk caps and inductors

F1C200s boot console / engineering prototype

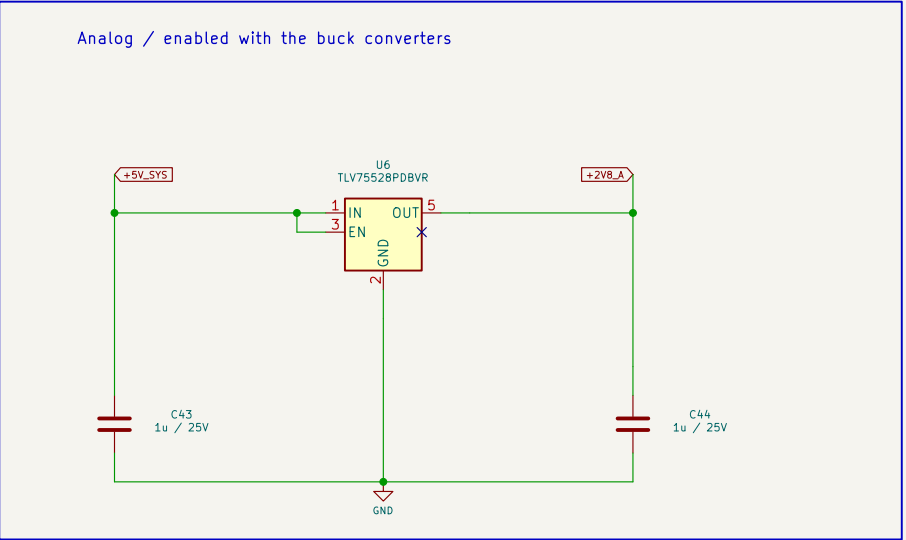
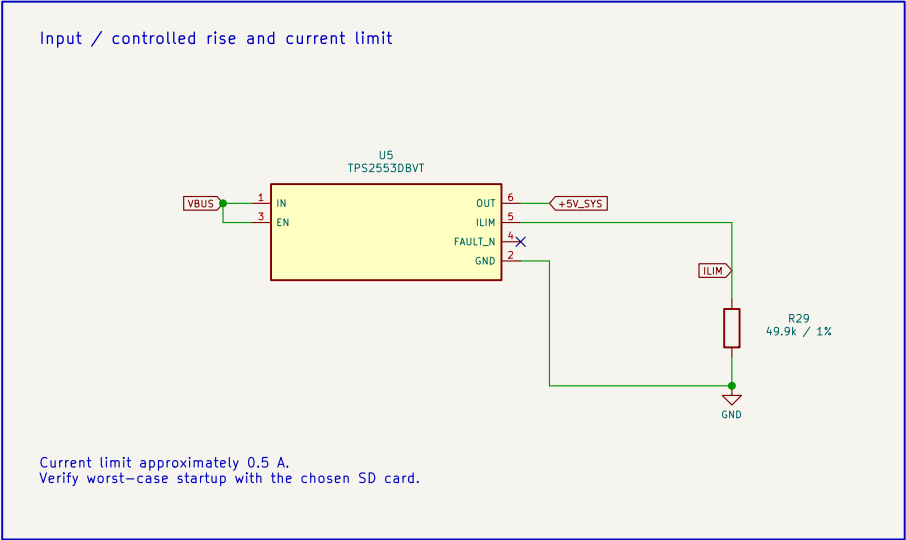
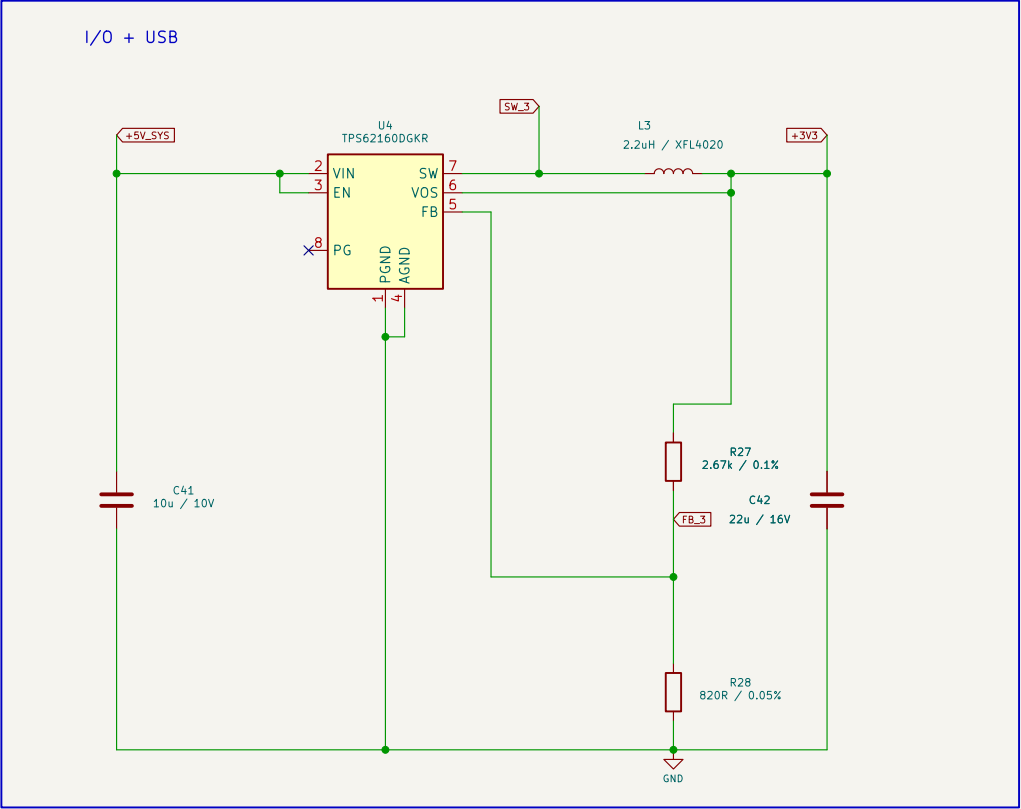
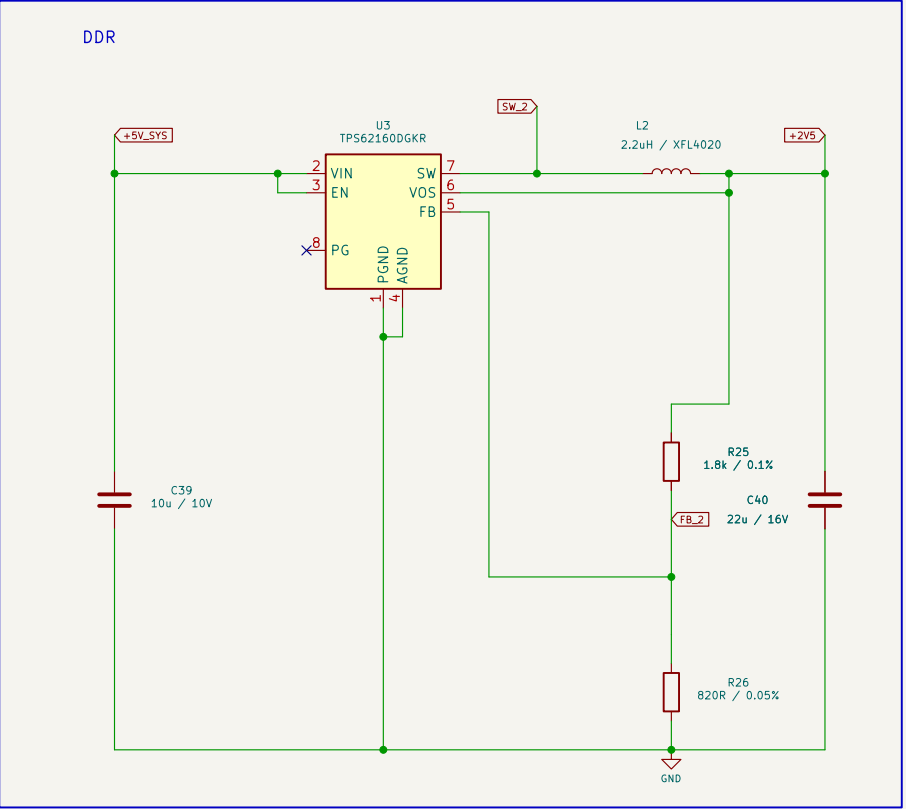
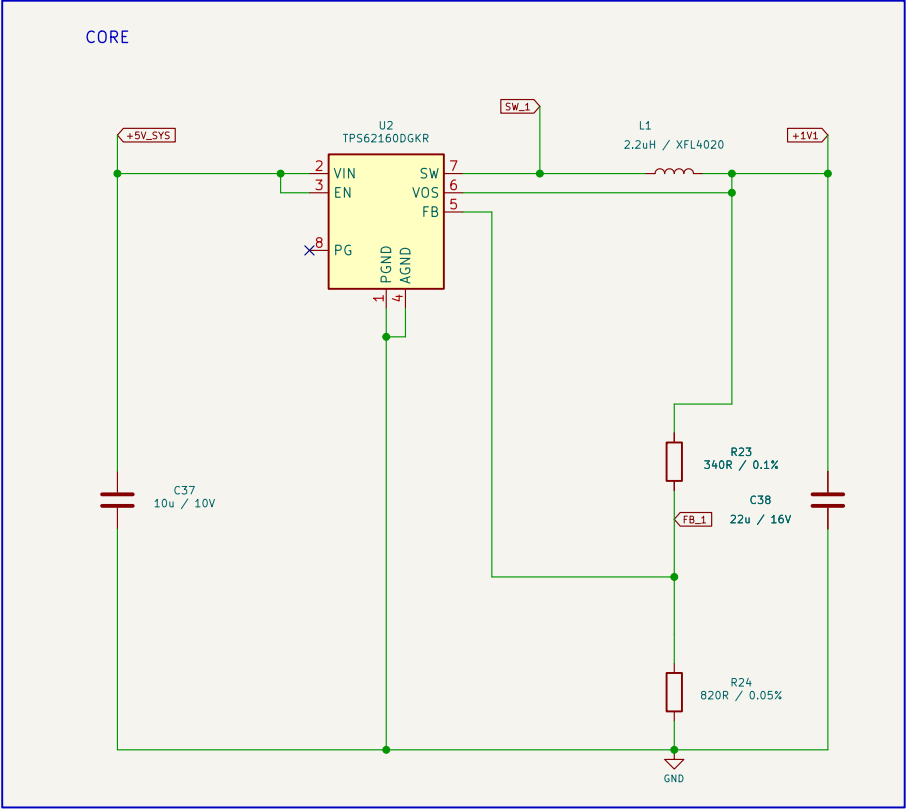
Sheet: /
File: boot-console.kicad_sch

Title: F1C200s / minimum Linux boot console

Size: A4	Date: 2026-09-06	Rev: 0.1
KiCad E.D.A. 10.0.4		Id: 1/4

Common-enable rails / reset waits for all supplies

Core 1.131707 V / DDR 2.556098 V / I/O 3.404878 V / analog 2.8 V



Startup and layout

All converters enable from protected 5 V.

U7 monitors all four rails; see sheet 03.

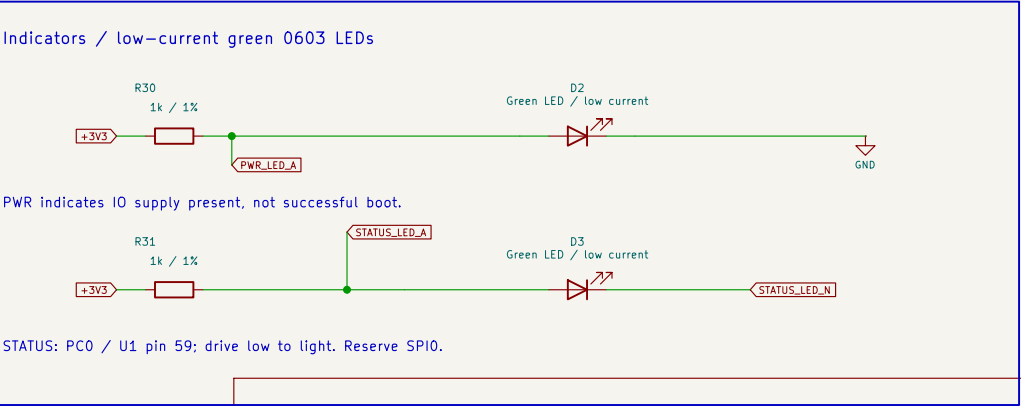
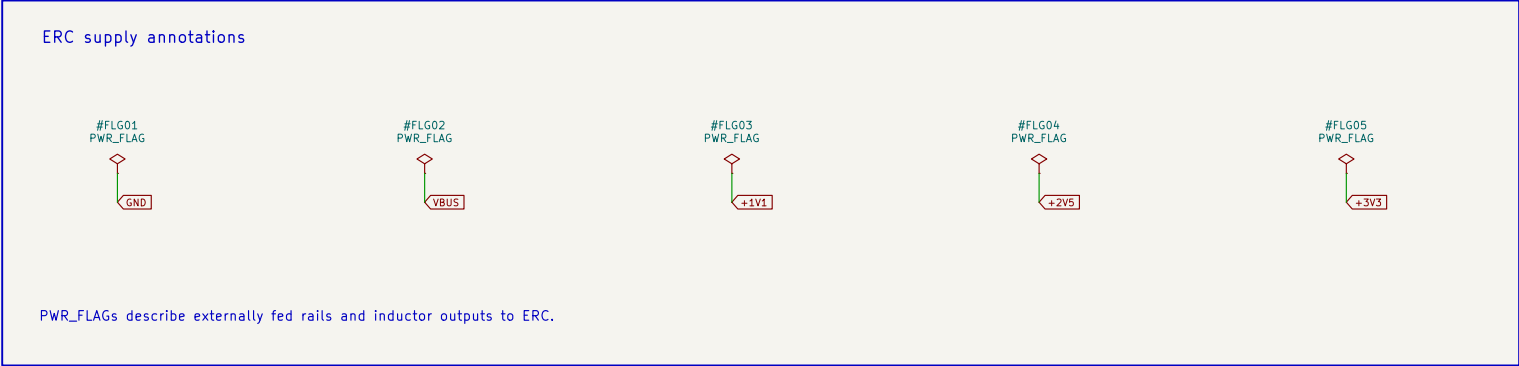
PG outputs are intentionally unused:

they float during UVLO.

Keep VIN loops and SW copper compact.

Return feedback dividers to local AGND.

Validate output capacitance at DC bias.



F1C200s boot console / engineering prototype

Sheet: /01 / Power and supervised startup/
File: power.kicad_sch

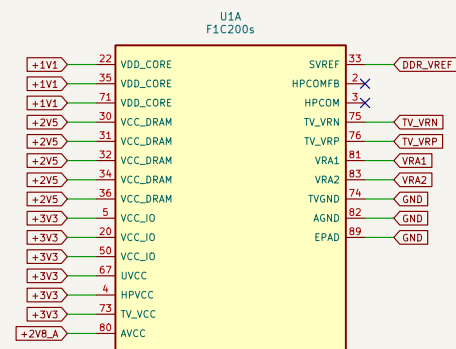
Title: 01 / Power and supervised startup

Size: A3 Date: 2026-09-06
KiCad E.D.A. 10.0.4

Rev: 0.1
Id: 2/4

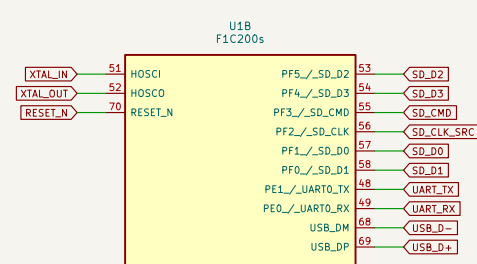
F1C200s / ARM9 + 64 MiB SiP DDR1

Processor supplies

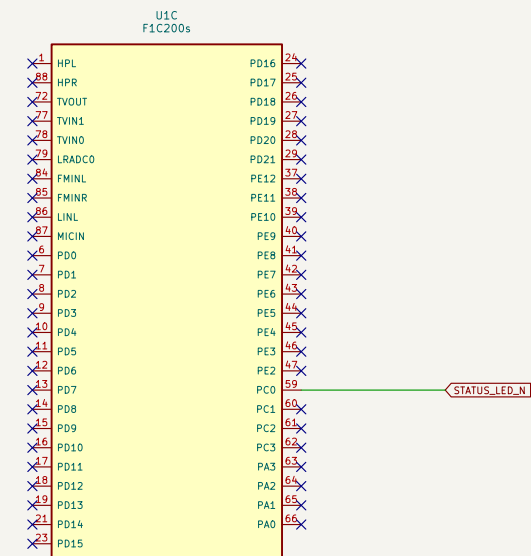


Connect EPAD and thermal vias to the solid ground plane.

Boot and console signals

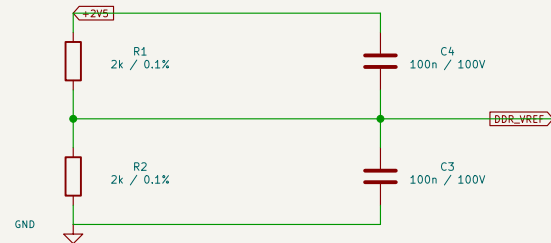


Spare pins and status output

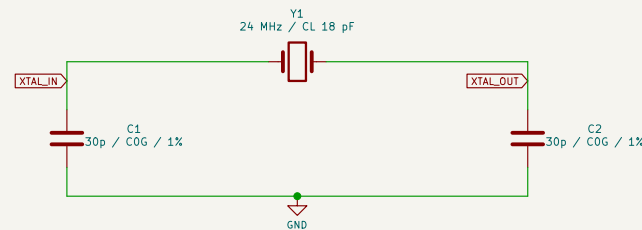


Spare GPIO and analog pins; PC0 drives the status LED.

DDR reference / matched divider / quiet local return

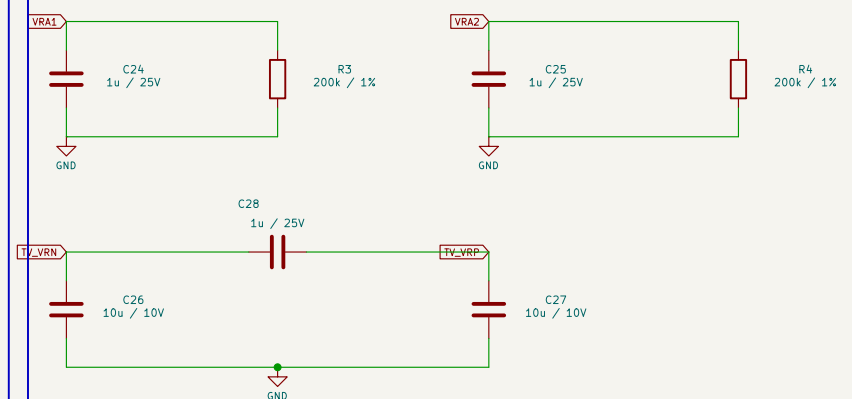


24 MHz / crystal and load capacitors

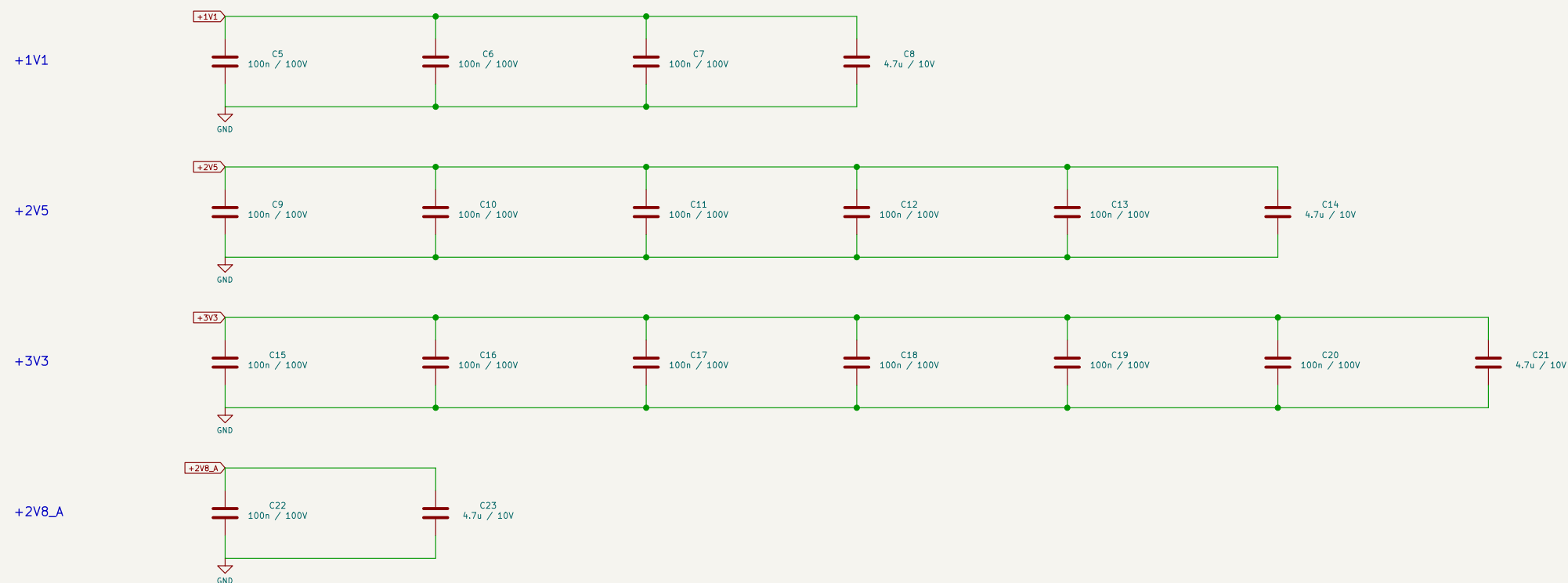


Place at pins 51 and 52. $CL = C/2 + C_{stray}$.
30 pF initial loads; tune after measurement.

Analog references / retained support



Local 0603 bypass / one 100 nF capacitor per supply pin



F1C200s boot console / engineering prototype

Sheet: /02 / F1C200s, clock and local bypass/
File: processor.kicad_sch

Title: 02 / F1C200s, clock and local bypass

Size: A3

Date: 2026-09-06

Rev: 0.1

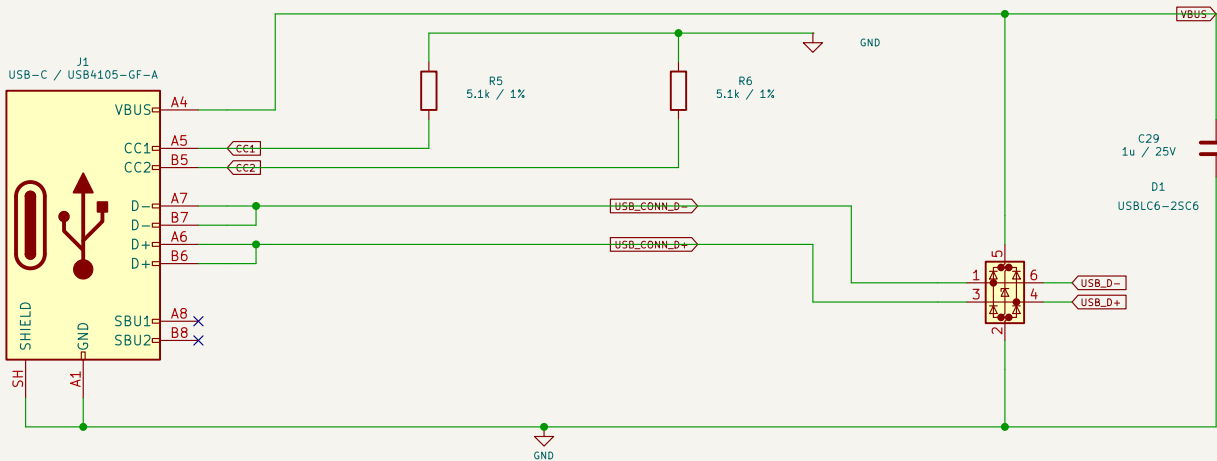
Size: A5	
KiCad E.D.A. 10.0.4	

Id: 3/4

USB-C / 5 V sink + native USB 2.0 recovery and gadget

microSD / removable boot image

USB input and ESD protection

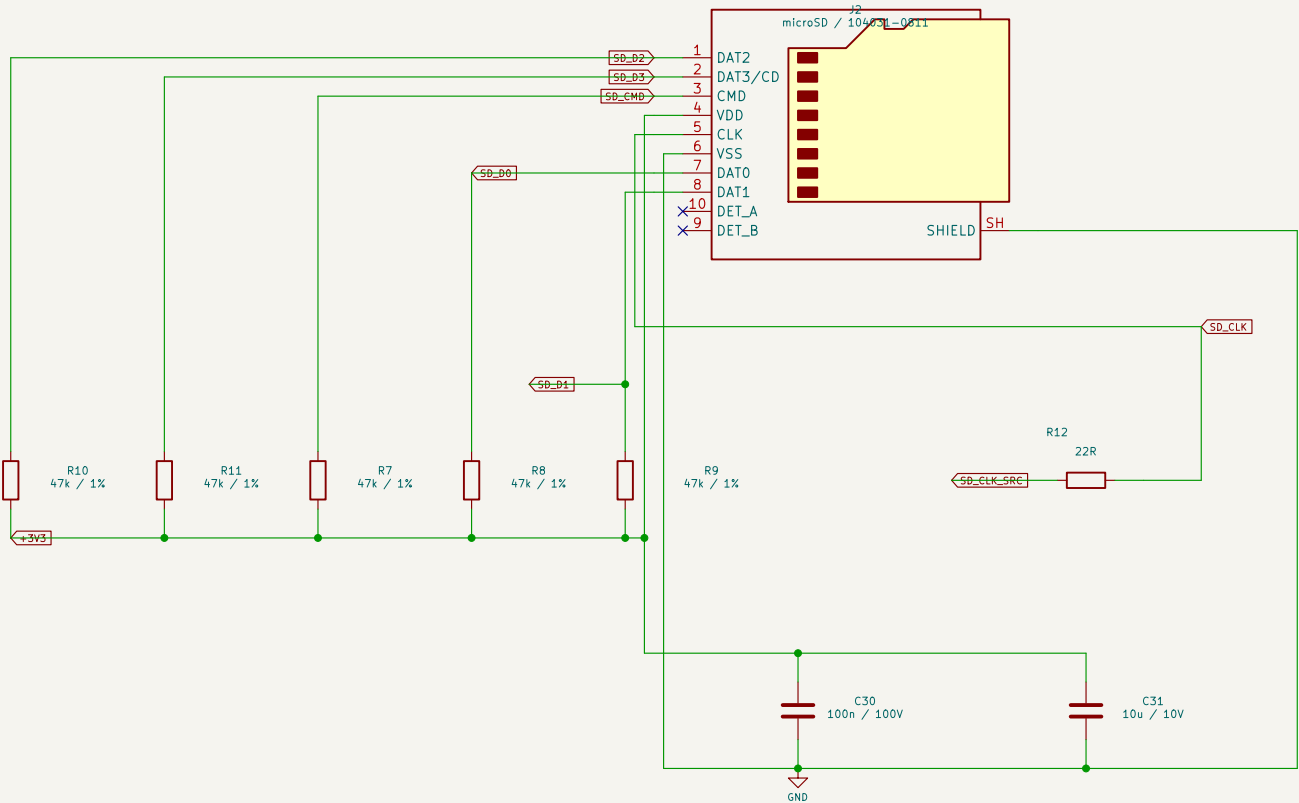


No USB-to-UART bridge. First-boot logs use J3.

USB_CONN_D+/D- enter D1; USB_D+/D- leave through its internal channels.

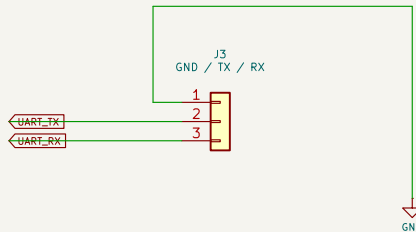
Use a 5 V USB supply for bring-up; no USB-PD or host power output.

microSD socket and bus support



Card-detect switch unused: use broken-cd; power off before card removal.

Console / UART0, 3.3 V logic, 115200 8N1

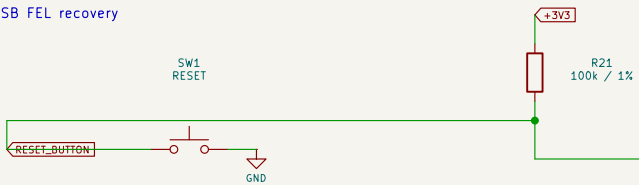


Adapter RX <- board TX; adapter TX -> board RX; common GND.

Do not connect an adapter power pin. Native RS-232 voltage is unsuitable.

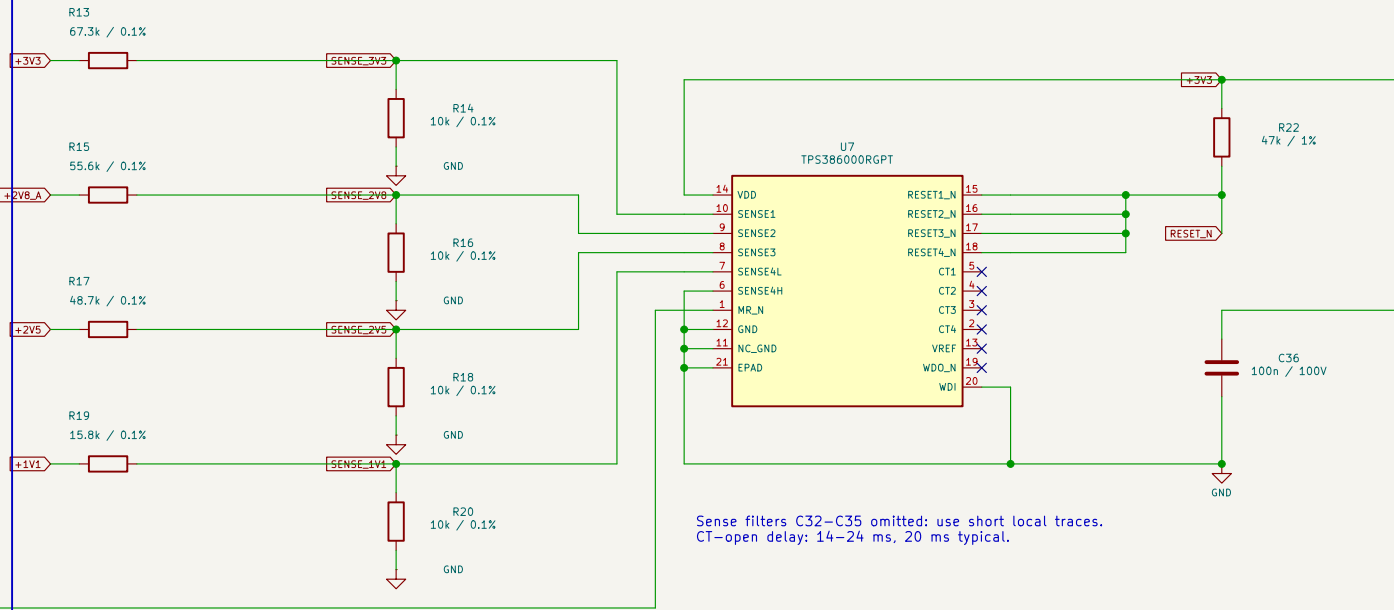
Reset / press to restart

No SD card: native USB FEL recovery



Reset remains asserted while any rail is not valid.

Four-rail reset supervisor / 14-24 ms delay



Sense filters C32-C35 omitted: use short local traces.
CT-open delay: 14-24 ms, 20 ms typical.

F1C200s boot console / engineering prototype

Sheet: /03 / USB-C, microSD and UART console/
File: interfaces.kicad_sch

Title: 03 / USB-C, microSD and UART console

Size: A3 Date: 2026-09-06
KiCad E.D.A. 10.0.4

Rev: 0.1
Id: 4/4